Significant figures, precision and accuracy: answers

Exercise 1

1 **a** 0.00472: 3 · 5003: 4 · 0.0480: 3 · 70.0: 3 · 2.50: 3 · 0.0905: 3 · 500.56: 5.
b No: the zeros at the end could be placeholders or measured, so it could have 2, 3 or 4 s.f.

2 a) 4.57 b) 0.0433 c) 0.900 d) 1.00 e) 45 700

3 a) 46 000 b) 0.074 c) 3.0 d) 0.061 e) 13

4 12.3456: 12.35 | 12 · 0.0605: 0.06 | 0.061 · 0.00093: 0.00 | 0.00093

5 0.0036 mm: 4 d.p., 2 s.f. · 3.6 µm: 1 d.p., 2 s.f. Significant figures do not change with the unit.

6 He started counting at the decimal point, not at the first non-zero digit (the 3). Correct: 0.035.

7 6.04 and 6.040. The zero is after the decimal point, so it counts as a significant figure: it says the thousandths were kept.

8 Any such number, for example 0.0035 or 0.0720.

Exercise 2

1 **a** P: both (mean 50.0 g, readings within 0.2 g). Q: precise, not accurate (mean about 53.4 g, too high). R: accurate on average (mean 50 g), not precise (4 g apart). S: neither (mean 53 g, readings 13 g apart).
b P: accurate and precise.
c 53 g, 54 g, 53 g. Still about 3 to 4 g too high, so not accurate. Rounding changed the digits, not the distance to 50 g.

2 **a** Yes: readings within 0.2 mm. **b** No: mean 4.1 mm, about 0.9 mm too thin. **c** Her cutting is consistent already; she should cut thicker slices.

3 **a** 58.08 ÷ 4 = 14.52 cm. **b** 14.58 − 14.47 = 0.11 cm. **c** 14.5 cm, 3 s.f.

4 **a** The ruler cannot show that much detail; many digits would claim a precision the measurement does not have (false precision). **b** No: rounding changes the precision we claim, not how close the answer is to the true length.

5 Mean = 300 ÷ 3 = 100 g. 100.000 g claims a precision of a thousandth of a gram, but the readings differ by 2 g (false precision). Write 100 g.

Challenge e.g. precise, not accurate: 24.1 g, 24.0 g, 24.1 g. Accurate on average, not precise: 17 g, 23 g, 20 g.

What do you think? 1. You cannot tell from one display: accuracy needs the true mass, precision needs repeated readings. 2. The digits tell you how much the display can show, not whether the reading is right or consistent.